

# ECG Monitoring System using AD8232 and Arduino UNO

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## 1. Abstract

This project presents the design and implementation of a low-cost ECG monitoring system using the AD8232 sensor and Arduino UNO. The system captures electrical signals from the human heart, processes them using digital filtering techniques, and displays the waveform using the Arduino Serial Plotter. The aim is to develop a simple and efficient real-time ECG monitoring solution.

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## 2. Introduction

Electrocardiography (ECG) is a technique used to measure the electrical activity of the heart. It is widely used for detecting heart conditions and monitoring cardiac health. This project focuses on building a cost-effective ECG monitoring system using AD8232 and Arduino UNO for real-time signal acquisition and visualization.

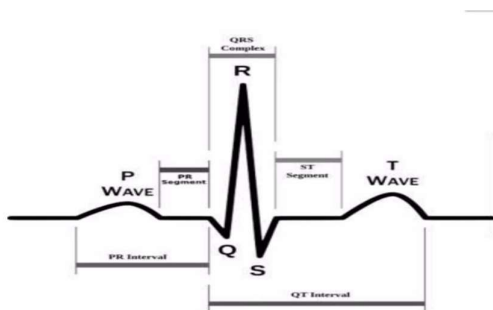


Fig 1: ECG Waveform(P-QRS-T Complex)

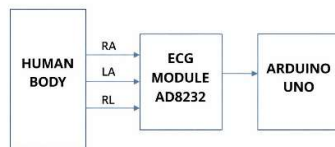
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## 3. Objectives

- To design a low-cost ECG monitoring system

- To acquire real-time ECG signals using AD8232
  - To process ECG signals using filtering techniques
  - To display ECG waveform using Arduino Serial Plotter
  - To ensure proper electrode detection using lead-off mechanism
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## 4. Block Diagram



### Explanation of Block Diagram

The ECG signal is collected from the human body using electrodes (RA, LA, RL).

The AD8232 module amplifies and filters the weak ECG signals.

The Arduino UNO reads the analog signal and converts it into digital form.

The processed signal is then displayed using Serial Plotter.

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## 5. Working model diagram:

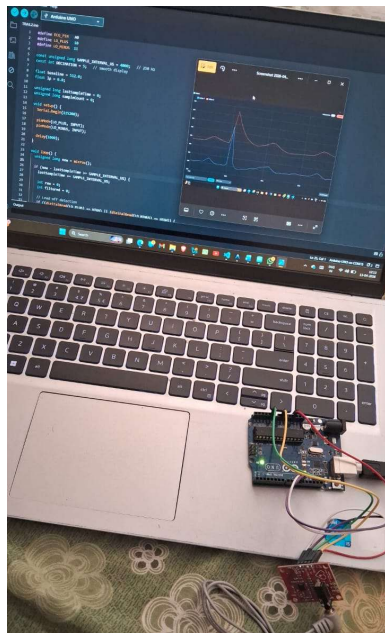


Fig 2: Working Model of ECG System

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## 6. Components Used

- Arduino UNO
  - AD8232 ECG Sensor
  - Electrodes (RA, LA, RL)
  - Jumper wires
  - Computer with Arduino IDE
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## 7. Prototype Design:

The prototype shows the connection of electrodes to the AD8232 ECG module and Arduino UNO.

The electrodes (RA, LA, RL) capture heart signals, which are amplified by the AD8232 module.

The processed signal is then sent to Arduino for further processing and display.

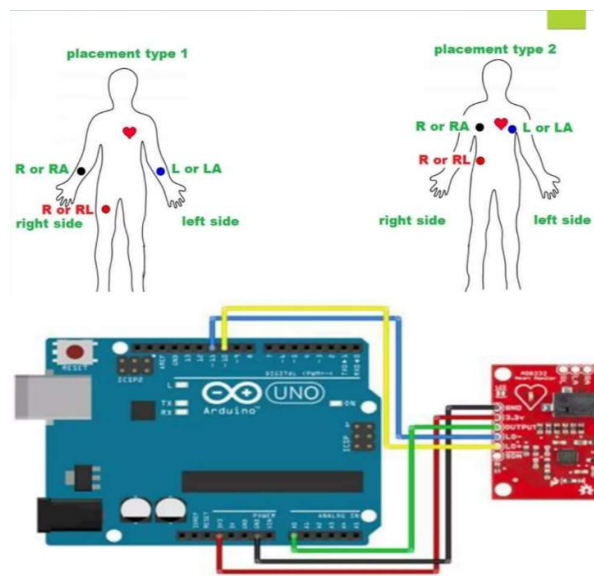


Fig 3: Prototype Design of ECG Monitoring System

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## 8. Working Principle

The Electrocardiogram (ECG) monitoring system works by capturing the electrical activity of the human heart using surface electrodes placed on the body. These electrodes (RA, LA, RL) detect very small bioelectric signals generated during cardiac activity.

The detected signals are extremely weak (in millivolts) and are therefore amplified using the AD8232 ECG module. This module also performs initial analog filtering to remove motion artifacts and unwanted noise.

The conditioned analog signal is then fed into the Arduino Uno through an analog input pin (A0). The Arduino converts this analog signal into digital form using its built-in Analog-to-Digital Converter (ADC).

Once digitized, the signal undergoes further processing using software-based filtering techniques such as baseline correction and low-pass filtering to improve signal quality. The processed ECG waveform is then transmitted via serial communication and displayed using the Arduino Serial Plotter for visualization of heart activity (P, QRS, T waves).

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## 9. Code Explanation

The implemented code performs real-time acquisition and processing of ECG signals using the Arduino platform.

### Sampling

The ECG signal is sampled at a rate of **250 Hz**, which is suitable for accurately capturing cardiac signals without losing important waveform details.

### Lead-Off Detection

The system continuously checks the status of electrode connections using LO+ and LO- pins. If electrodes are disconnected, the system outputs zero values to avoid displaying incorrect data.

### Baseline Removal

To eliminate baseline drift caused by breathing and body movement, an exponential moving average method is used. This ensures that the ECG waveform remains centered and stable.

### Low-Pass Filtering

A digital low-pass filter is applied to remove high-frequency noise such as power line interference and muscle artifacts. This helps in preserving the essential features of the ECG signal, especially the QRS complex.

### Signal Amplification and Offset

The filtered signal is amplified to enhance visibility and shifted to fit within the ADC range (0–1023), making it suitable for plotting.

### Decimation (Data Reduction)

To ensure smooth visualization and reduce data overload, the system outputs only every nth sample. This improves readability in the Serial Plotter.

## Output Visualization

Finally, both raw and filtered ECG signals are transmitted via serial communication and displayed graphically using the Arduino Serial Plotter

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## 10. Results



The system successfully displays ECG waveform including P, QRS, and T waves. The filtered signal is smoother and clearer compared to raw signal.

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## 11. Conclusion

The ECG monitoring system successfully captures and displays real-time heart signals. Digital filtering improves signal quality by reducing noise and baseline drift. This system provides a low-cost solution for basic ECG monitoring and can be further enhanced for advanced applications.